



Suspect zero

The first ever mobile 'virus' was in fact an anti-piracy code made by a company called Ojam in its Mosquito game back in 2004, to get back at illegal users



Botnets

A collection of devices infected on an always-on network set up to forward transmissions including spam or viruses to other devices

Siddharth Parwatay

siddharth.parwatay@thinkdigit.com

Even the most cautious of PC users will have certainly faced at least one instance of virus infection. We all know how frustrating a virus attack can be. Viruses, worms and trojans, under whatever category they're classified, can cause anywhere from minor irritation to total system failure. Instances of PC virus attacks are fairly common. These days, with mobile devices becoming increasingly similar to fully functional PCs, it makes one wonder whether

The virus story so far...

Ever since the iPhone changed our view of smartphones, there has been no stopping this segment. But that doesn't mean that smartphones weren't around prior to that. Sometime in 2004 came devices that were able to download applications from the internet and share software with one another through short-range Bluetooth connections; they also had worldwide multimedia messaging service (MMS) communications and memory cards. With these phones came the first mobile phone virus written for the Symbian series 60 OS. It was dubbed

The later variants of Cabir and other viruses could completely disable a phone, delete the data on it, turn icons into skulls, or force the device to send costly messages to premium-priced numbers. Soon the number of recorded viruses for mobile phones grew to about a hundred, but mostly on Symbian and Windows Mobile platforms. This growth was very similar to the growth rates of viruses on the PC platform after Brain, the first pc virus, made its appearance.

Soon, by 2006, there were several Trojans doing the rounds. Some of the common mobile Trojans to have attacked mobile

All you ever wanted to know about mobile phone viruses and other malware, and how you can protect yourself

The stage is set

Enter the mobile virus



these devices will be the next to face an epidemic of a virus threat. Turns out they are surely on the target list for malicious code programmers and that we may be on the brink of a widespread virus attack in the near future. A recent research study that appeared in *Science* magazine had scientists studying human behavioural patterns to find out the reasons behind this and what to expect if such a scenario were to take place. The study revealed that the only type of phone that is susceptible to a virus attack is one that runs an operating system – a smartphone. Considering that even in these difficult times the growth of smartphones is unstoppable and nearly doubling every year, the time is not far off when we see a full-fledged virus attack.

Cabir and was a classic proof-of-concept virus, clearly created to prove a point and perhaps claim bragging rights. The first variant of Cabir was not very harmful; it simply replicated itself through Bluetooth and draining the battery was its only malicious intent. Later variants, however, were not so benign and did considerable damage. The transmission mode for Cabir was through Bluetooth. The virus would basically search for other devices that were left in the 'discoverable' mode and jump that way from phone to phone. Cabir comes into a phone as a Bluetooth message in the inbox as a caribe.sis file. Opening the file would unleash the virus and soon it starts looking for other devices.

Soon many malicious code writers took a cue from Cabir and began modifying it to make more damaging forms of the virus.

phones and smart phones are: SymbOS. Sendtool – a trojan horse that runs on the Symbian operating system. This trojan drops a hacktool that can be used to send malicious programs, such as variants of the SymbOS.PBStealer family of trojans, to other mobile devices via Bluetooth. CDropper is another family of Symbian SIS file trojans that will install Cabir variants into the device. Mainly, this variant makes the menu icons go blank. Commwarrior then emerged on the scene and had its biggest impact in Spain where it infected about 1,00,000 phones in 2006. It infected phones running Nokia's Symbian series 60 second edition operating system and spread via MMS messages – text messages containing an audio, video or picture file. Certainly an irritant, it drew up large phone bills for users. According to Panda Security,